



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Practical 6

AIM:- Use string functions and regex for email validation , data extraction , and text analysis

String & Regex Analyzer

Validate formatting, run text extraction rules, and parse analytics instantly.

Live Email Validation

sanvi.wadhankar.batch2024@sitnagpur.siu.edu.in

✓ sanvi.wadhankar.batch2024@sitnagpur.siu.edu.in

Unstructured Source Text

[Load Sample Text](#)

For database architecture issues, ping sanvi wadhankar at sanvi.wadhankar.batch2024@sitnagpur.siu.edu.in or call 800-555-0199.

The backup service email is backup-service@cloud-infra.net. We also found an old file log from 2024-12-01. Please call the alternate helpline 555-123-4567 if the primary line is busy. Teamwork makes the dream work!

Process & Analyze Text

Extracted Patterns

DETECTED EMAILS

sanvi.wadhankar.batch2024@sitnagpur.siu.edu.in

DETECTED PHONE NUMBERS

800-555-0199, 555-123-4567

DETECTED DATES (YYYY-MM-DD)

2026-09-15, 2024-12-01

Text Metrics Dashboard

Total Words

53

Unique Words

47

TOP 5 MOST FREQUENT WORDS

"the"	4 instances
"please"	2 instances
"call"	2 instances
"is"	2 instances
"20241201"	1 instances

Created by Sanvi Wadhankar

What are String Functions?

String functions are built-in actions you can perform on text data (strings). Because strings in JavaScript are immutable, these functions never change the original text; instead, they return a brand-new string or value

Common String Functions:

- slice(start, end): Extracts a piece of a string.
- toLowerCase() / toUpperCase(): Changes the text casing.
- trim(): Strips out whitespace from both ends of the text.

What is Regex (Regular Expressions):-

A **Regex** is a special sequence of characters that forms a **search pattern**. Think of it as a superpowered "Find and Replace" tool. You can use it to check if an email address is valid, find phone numbers in a block of text, or replace formatting

Task	Core Method	Recommended Regex Pattern	What it Does
Email Validation	regex.test(string)	/^[^\s@]+@[^\s@]+\.[^\s@]+\$/	Returns true or false based on format.
Data Extraction	string.match(regex)	/[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}/g	Pulls all emails into an array.

Conclusion:-

Hence we have performed email vaidation and data extraction , text analysis using regular expressions and string functions